

Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Computo en la nube

4.3 Seguridad (hardening) en servidor web Apache 2

Docente: Ing. Miguel Salazar del Bosque

Equipo:

Hugo Emilio Espinoza Tun 22050627

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Paso 1: Instalación del servidor Apache

```
hugo@debian:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; pres
   Active: active (running) since Fri 2026-03-20 08:21:12 CDT; 7s ago
   Invocation: 0fdfbf0dedf24b368ed02b559bd43ffc
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 1248 (apache2)
      Tasks: 6 (limit: 2301)
     Memory: 12.2M (peak: 12.2M)
        CPU: 50ms
    CGroup: /system.slice/apache2.service
            └─1248 /usr/sbin/apache2 -k start
              └─1251 /usr/sbin/apache2 -k start
                └─1252 /usr/sbin/apache2 -k start
                  └─1253 /usr/sbin/apache2 -k start
                    └─1254 /usr/sbin/apache2 -k start
                      └─1255 /usr/sbin/apache2 -k start

Mar 20 08:21:12 debian systemd[1]: Starting apache2.service - The Apache HT
Mar 20 08:21:12 debian systemd[1]: Started apache2.service - The Apache HT
hugo@debian:~$
```



Paso 2: Verificar funcionamiento en el navegador

No seguro 192.168.56.20



Apache2 Debian Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/  
|-- apache2.conf  
|   |-- ports.conf  
|-- mods-enabled  
|   |-- *.load  
|   |-- *.conf  
|-- conf-enabled  
|   |-- *.conf  
|-- sites-enabled  
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or

Paso 3: Ocultar información sensible

```
hugo@debian: /v × hugo@debian: /v × hugo@debian: /v × + v
GNU nano 8.4 /etc/apache2/conf-available/security.conf
# Changing the following options will not really affect the security of the
# server, but might make attacks slightly more difficult in some cases.

#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
#ServerTokens OS
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
ServerSignature Off
#ServerSignature On

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On

#
# Forbid access to version control directories
#
# If you use version control systems in your document root, you should
[ Read 59 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify

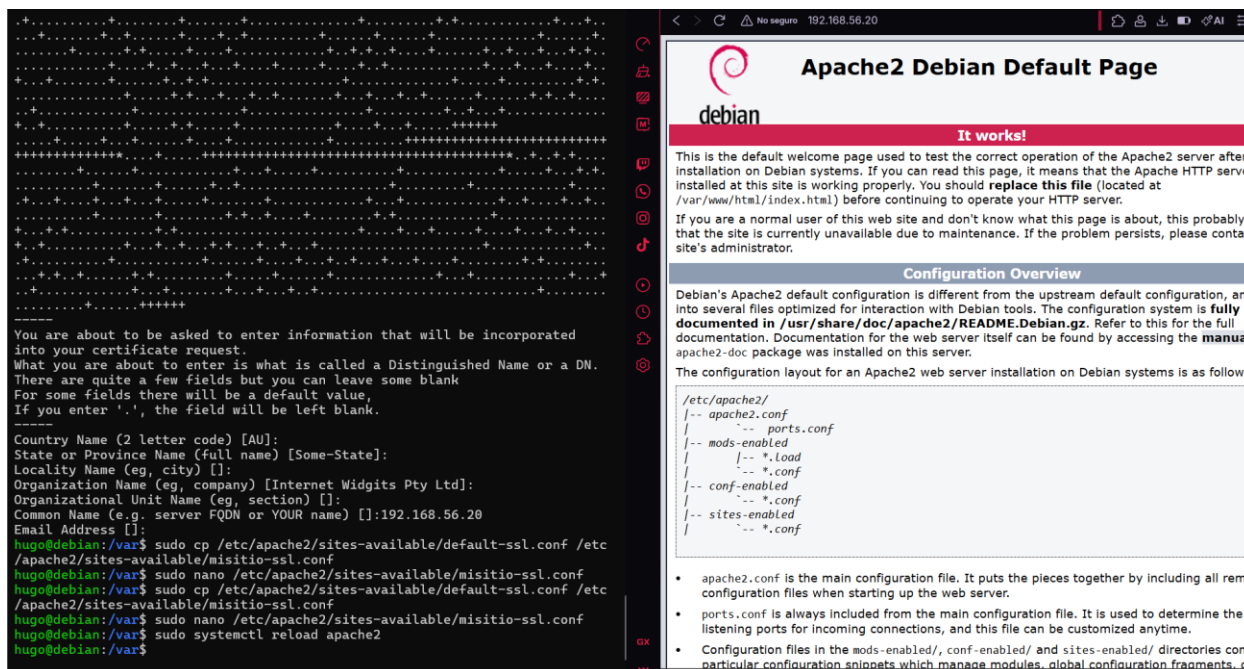
hugo@debian:/var$ sudo a2enconf security
sudo systemctl restart apache2
Conf security already enabled
hugo@debian:/var$ |
```



Paso 4: Eliminar módulos innecesarios

```
hugo@debian:/var$ sudo a2dismod autoindex status proxy proxy_http
WARNING: The following essential module will be disabled.
This might result in unexpected behavior and should NOT be done
unless you know exactly what you are doing!
autoindex

To continue type in the phrase 'Yes, do as I say!' or retry by passing '-f':
Yes, do as I say!
Module autoindex disabled.
Module status disabled.
Module proxy already disabled
Module proxy_http already disabled
To activate the new configuration, you need to run:
systemctl restart apache2
hugo@debian:/var$ sudo systemctl restart apache2
hugo@debian:/var$ |
```



Paso 6: Firewall y puertos

```
hugo@debian:/var$ sudo ss -tuln
Netid  State  Recv-Q  Send-Q               Peer Address:Port              Local Address:Port
udp    UNCONN  0        0               0.0.0.0:*          10.0.2.15:68
udp    UNCONN  0        0       [fd17:625c:f037:2:a00:27ff:fe3b:1113]:546
      [::]:*
udp    UNCONN  0        0       [fd17:625c:f037:2:3ba4:89d4:4fbb:ab35]:546
      [::]:*
udp    UNCONN  0        0       [fe80::3919:48eb:eebc:752d]%enp0s3:546
      [::]:*
tcp    LISTEN  0       128               0.0.0.0:*          0.0.0.0:22
tcp    LISTEN  0        80               0.0.0.0:*          127.0.0.1:3306
tcp    LISTEN  0       128               0.0.0.0:*          [::]:22
tcp    LISTEN  0       511               *:*:               *:80
tcp    LISTEN  0       511               *:*:               *:443
      *:*
```

```
hugo@debian:/var$ sudo ufw allow 80
Rule updated
Rule updated (v6)
hugo@debian:/var$ sudo ufw allow 443
sudo: ufw: command not found
hugo@debian:/var$ sudo ufw allow 443
Rule added
Rule added (v6)
hugo@debian:/var$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)?
y
Firewall is active and enabled on system startup
hugo@debian:/var$ |
```

Paso 7: Revisar logs y monitoreo

```
hugo@debian:/var$ sudo tail -f /var/log/apache2/access.log
192.168.56.1 - - [20/Mar/2026:08:46:41 -0500] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 OPR/128.0.0.0"
192.168.56.1 - - [20/Mar/2026:08:46:41 -0500] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://192.168.56.20/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 OPR/128.0.0.0"
192.168.56.1 - - [20/Mar/2026:08:48:42 -0500] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
192.168.56.1 - - [20/Mar/2026:08:48:42 -0500] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://192.168.56.20/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
192.168.56.1 - - [20/Mar/2026:08:49:34 -0500] "-" 408 0 "-" "-"
192.168.56.1 - - [20/Mar/2026:09:04:14 -0500] "GET / HTTP/1.1" 200 3367 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 OPR/128.0.0.0"
192.168.56.1 - - [20/Mar/2026:09:04:28 -0500] "GET / HTTP/1.1" 200 3367 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36 OPR/128.0.0.0"
^C
hugo@debian:/var$ sudo tail -f /var/log/apache2/error.log
[Fri Mar 20 08:55:03.454967 2026] [core:notice] [pid 1914:tid 1914] AH00094: Command line: '/usr/sbin/apache2'
[Fri Mar 20 08:56:20.044047 2026] [mpm_prefork:notice] [pid 1914:tid 1914] AH00170: caught SIGWINCH, shutting down gracefully
[Fri Mar 20 08:56:20.184993 2026] [mpm_prefork:notice] [pid 2021:tid 2021] AH00163: Apache/2.4.66 (Debian) OpenSSL/3.5.5 configured -- resuming normal operations
[Fri Mar 20 08:56:20.185053 2026] [core:notice] [pid 2021:tid 2021] AH00094: Command line: '/usr/sbin/apache2'
[Fri Mar 20 08:58:15.958686 2026] [mpm_prefork:notice] [pid 2021:tid 2021] AH00170: caught SIGWINCH, shutting down gracefully
[Fri Mar 20 08:58:16.279857 2026] [mpm_prefork:notice] [pid 2104:tid 2104] AH00163: Apache/2.4.66 (Debian) OpenSSL/3.5.5 configured -- resuming normal operations
[Fri Mar 20 08:58:16.279957 2026] [core:notice] [pid 2104:tid 2104] AH00094: Command line: '/usr/sbin/apache2'
[Fri Mar 20 09:04:09.618970 2026] [mpm_prefork:notice] [pid 2104:tid 2104] AH00171: Graceful restart requested, doing restart
[Fri Mar 20 09:04:09.695882 2026] [mpm_prefork:notice] [pid 2104:tid 2104] AH00163: Apache/2.4.66 (Debian) OpenSSL/3.5.5 configured -- resuming normal operations
[Fri Mar 20 09:04:09.695902 2026] [core:notice] [pid 2104:tid 2104] AH00094: Command line: '/usr/sbin/apache2'
```

Paso 8: Checklist de Verificación

Verificación	Estado
Apache instalado correctamente	✓
Página por defecto visible	✓
Información de versión oculta	✓
Módulos innecesarios deshabilitados	✓
HTTPS activo con certificado autofirmado	✓
Firewall limitado a puertos 80/443	✓
Logs activos y funcionando	✓

Conclusiones

La práctica realizada permitió comprender la importancia del hardening en servidores web, aplicando medidas de seguridad como la ocultación de información sensible, la desactivación de módulos innecesarios, la configuración de HTTPS y el control de accesos mediante firewall.

Estas acciones contribuyen a reducir la superficie de ataque y dificultan posibles intentos de explotación, incluyendo escenarios relacionados con la escalada de privilegios, ya que un servidor mal configurado puede exponer información o servicios que faciliten accesos no autorizados.

En este caso, se logró implementar correctamente las configuraciones solicitadas, cumpliendo con los objetivos de la práctica y reforzando la seguridad del servidor Apache.